

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL1, BL2)	Assessment:	Assessment Tool 2 – Case Study Analysis
Weightage:	20% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems.(BL1, BL2)</i>		
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Section / Batch:	CSE-AI	Date:	02-08-2026

Instructions to Students

- Answer any ONE questions.
- Total 100 marks
- Include architecture, explanation, suitable Python/API approach, advantages, limitations and evaluation.
- Time allotted: 30 mins

Code of Conduct

I N Goutham(192472024)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N Goutham

SECTION A – Case Study Analysis**Q5. Case Study 5**

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation approach, expected outcome, and limitations.

Topic: LoRA fine-tuning

Developing a Domain-Specific Medical AI Assistant using LoRA Fine-Tuning**1. Introduction**

Large Language Models (LLMs) such as GPT are trained on massive datasets and possess strong general language understanding. However, they often lack expertise in specialized domains such as healthcare, law, finance, or education. Retraining an entire LLM for every new domain is computationally expensive, requiring billions of trainable parameters, high-end GPUs, and enormous datasets.

Low-Rank Adaptation (LoRA) is a Parameter-Efficient Fine-Tuning (PEFT) technique that enables efficient adaptation of pretrained LLMs by training only a small number of additional parameters while

keeping the original model frozen. This significantly reduces memory usage, training time, and computational cost without sacrificing performance.

In this case study, LoRA is applied to fine-tune a pretrained GPT model for developing a medical question-answering assistant capable of providing accurate healthcare information.

2. Problem Statement

Hospitals and healthcare organizations require intelligent virtual assistants capable of answering medical questions, explaining symptoms, assisting doctors with documentation, and educating patients.

General-purpose LLMs often produce:

- Generic responses
- Medical inaccuracies (hallucinations)
- Lack of domain-specific terminology
- Inconsistent recommendations

Training a completely new model is expensive and impractical.

Therefore, an efficient fine-tuning technique is required.

3. Analysis of the Problem

Healthcare information constantly evolves with new diseases, treatment protocols, and medical research. Updating a full LLM every time new knowledge becomes available is computationally expensive.

Challenges include:

- Large model size
- High GPU memory requirements
- Long training time
- Expensive cloud infrastructure
- Need for specialized medical knowledge

LoRA addresses these challenges by introducing trainable low-rank matrices while freezing the pretrained weights.

This enables rapid adaptation to new domains using limited computational resources.

4. Justification for Choosing LoRA

LoRA is selected because it offers the best balance between accuracy, efficiency, and cost.

Advantages

- Only a small number of parameters are trained.
- Original pretrained weights remain unchanged.
- Reduces GPU memory usage.
- Faster fine-tuning.

- Easy deployment.
- Multiple LoRA adapters can be created for different domains.
- Maintains performance comparable to full fine-tuning.

Compared to full fine-tuning, LoRA requires only a fraction of trainable parameters while delivering similar task performance.

5. Suitable Model Selection

Technique	Purpose	Suitability
GPT	Base Large Language Model	✓ Selected
BERT	Classification & embeddings	✗ Not suitable for text generation
RLHF	Aligns model responses with human preferences	Can be applied after LoRA
PEFT	Parameter Efficient Fine-Tuning framework	✓ Used
LoRA	Efficient domain adaptation	✓ Best choice

Selected Architecture

Base Model

GPT (e.g., GPT-2, Llama, Mistral)



PEFT Framework



LoRA Adapter



Medical Dataset



Fine-Tuned Medical Assistant

6. Why GPT Instead of BERT?

GPT	BERT
Generates text	Understands text
Autoregressive	Bidirectional encoder
Suitable for chatbots	Suitable for classification
Better for question answering and conversation	Better for sentiment analysis

Since the objective is generating medical responses, GPT is more appropriate.

7. Implementation Approach

Step 1: Select Pretrained Model

Choose a pretrained GPT model.

Example:

- GPT-2
- Llama 3
- Mistral

Step 2: Prepare Dataset

Collect domain-specific medical datasets.

Examples:

- Medical question-answer pairs
- Clinical notes
- Disease descriptions
- Treatment guidelines

Step 3: Tokenization

Convert medical text into tokens using the tokenizer corresponding to the selected GPT model.

Step 4: Apply LoRA

Freeze all pretrained model weights.

Insert trainable low-rank matrices into attention layers.

Instead of updating billions of parameters, LoRA updates only a few million parameters.

Step 5: Fine-Tune

Train using:

- Learning Rate
- Batch Size
- Epochs
- Validation dataset

Monitor:

- Training loss
- Validation loss
- Accuracy

Step 6: Evaluate

Evaluate using:

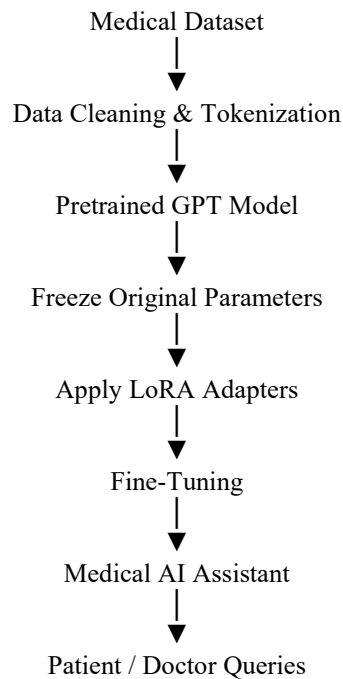
- BLEU Score
- ROUGE Score
- Perplexity
- Human evaluation
- Medical expert validation

Step 7: Deploy

Deploy as:

- Hospital chatbot
- Clinical decision support
- Medical FAQ assistant
- Patient support system

8. System Architecture



9. Expected Outcome

The fine-tuned system is expected to:

- Generate medically relevant responses.
- Reduce hallucinations compared to a general GPT model.
- Improve domain-specific accuracy.
- Require significantly less GPU memory.
- Reduce training time.
- Lower deployment cost.
- Enable easy updates using new LoRA adapters.

10. Real-World Implications

LoRA fine-tuning has significant real-world impact across multiple domains:

Healthcare

- Clinical decision support
- Patient education
- Medical chatbots
- Electronic health record assistance

Education

- Personalized tutoring
- Question answering

Finance

- Investment advisory
- Fraud detection assistants

Legal

- Legal document drafting
- Contract summarization

Customer Support

- Domain-specific virtual assistants

LoRA enables organizations to customize LLMs without retraining massive models, making advanced AI accessible even with limited computational resources.

11. Limitations

Although LoRA is highly efficient, it has certain limitations:

- Performance depends on the quality of domain-specific data.
- May still produce hallucinations.
- Cannot completely replace full fine-tuning for highly specialized tasks.
- Requires careful hyperparameter tuning.
- Medical outputs must be validated by experts.
- Ethical and privacy concerns remain.

12. Comparison: Full Fine-Tuning vs LoRA

Feature	Full Fine-Tuning	LoRA Fine-Tuning
Trainable Parameters	Billions	Millions
GPU Memory	Very High	Low
Training Cost	Very High	Low
Training Time	Long	Short

Feature	Full Fine-Tuning	LoRA Fine-Tuning
Storage	Large	Small
Performance	Excellent	Near Excellent
Deployment	Difficult	Easy

13. Conclusion

LoRA fine-tuning provides an efficient and scalable approach for adapting pretrained LLMs to domain-specific tasks. In this case study, LoRA enables a GPT-based medical assistant to deliver accurate and context-aware healthcare responses while significantly reducing computational requirements. Compared to full fine-tuning, LoRA achieves similar performance with lower memory usage, reduced training time, and lower cost. Combined with the PEFT framework and validated by medical experts, LoRA offers a practical solution for deploying reliable AI systems in healthcare and other specialized domains.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Depth	30%	Deep, accurate understanding of pre-training/fine-tuning/RLHF concepts	Good understanding with minor gaps	Superficial understanding of concepts	Misunderstands core concepts
Real-World Implications	25%	Draws well-reasoned real-world implications and comparisons	Reasonable implications drawn	Weak or generic implications	No meaningful analysis presented
Analytical Rigor	20%	Analysis is thorough, evidence-based, and well-structured	Mostly thorough analysis	Partial analysis with gaps in evidence	Superficial, unstructured analysis
Report Structure	15%	Report/slides professionally structured, easy to follow	Clear structure with minor issues	Structure unclear in places	Poorly structured submission
Citation & Academic Writing	10%	Properly cites sources; clear academic writing	Mostly cited, minor lapses	Inconsistent citation practice	No citation or referencing

Marks Evaluation:

Question No.	Conceptual Depth (30%)	Real-World Implications (25%)	Analytical Rigor (20%)	Report Structure (15%)	Citation & Academic Writing (10%)	100
Q1						
Overall Total (100)						